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Chasing the Nines
Reliable RL Across States and Seeds

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PENDULUM · 2,501 STARTS · FIVE INDEPENDENTLY TRAINED ACTORS

RL + supervised has 9 near-reference failures versus 497 for pure RL; pure RL wins strictly more often.

RL + supervised
9 near-reference failures
Task 93.858% · strict wins 12.555%
Near-reference **99.928%**

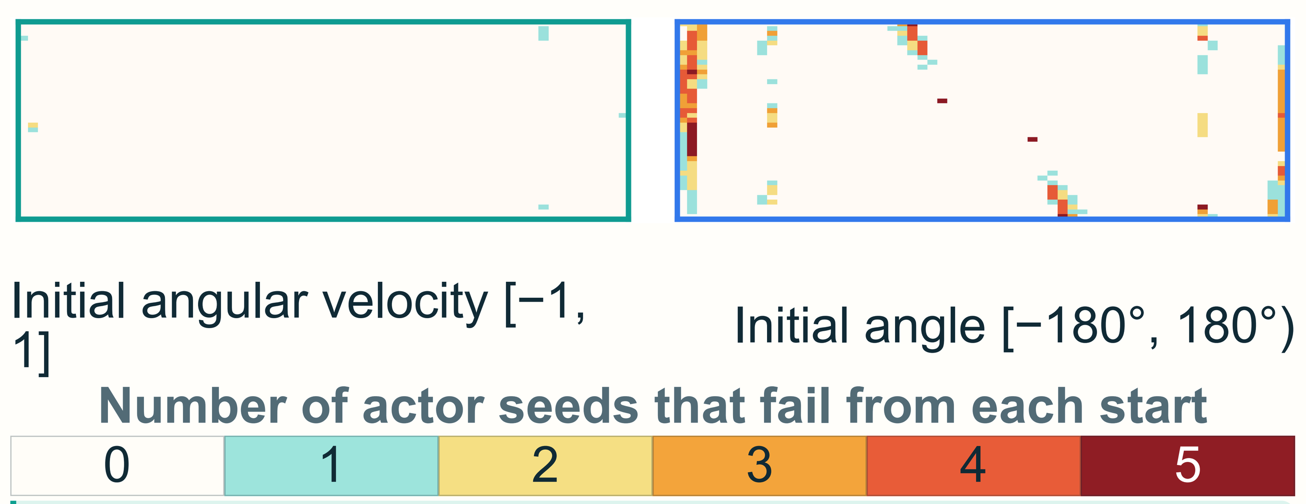
Pure RL
497 near-reference failures
Task 92.875% · strict wins 22.823%
Near-reference **96.026%**

01 Reliability is a tail problem

497 pure-RL near-reference failures **VS** **9** RL + supervised near-reference failures

Five actors × 2,501 starts. Each heatmap cell counts actor seeds below the near-reference threshold.

RL + supervised — 9 misses on 8 starts
Pure RL — 497 misses on 208 starts



All five actors pass 2,493 RL + supervised starts and 2,293 pure-RL starts. Pure-RL seeds disagree on 195 starts; all five fail on 13.

Six policy families on the same grid

Policy family	Near	Task	Strict
RL + supervised	99.93	93.86	12.55
Uniform-start mixed	99.85	93.84	11.67
Supervised actor	99.72	93.60	14.50
Selected pure RL	96.03	92.87	22.82
Plain SimbaV2	91.84	91.46	8.52
Clean DAgger	84.53	88.02	6.39

Protocol: 61 angles × 41 angular velocities, 200-step rollouts. **Near:** within 5 return of the better controller. **Task:** upright and slow for ≥80%. **Strict:** beats both controllers. Five seeds, 12,505 trials.

02 Two training routes, one fixed-grid scorecard

RL + supervised — 30k + 20k + 50k updates

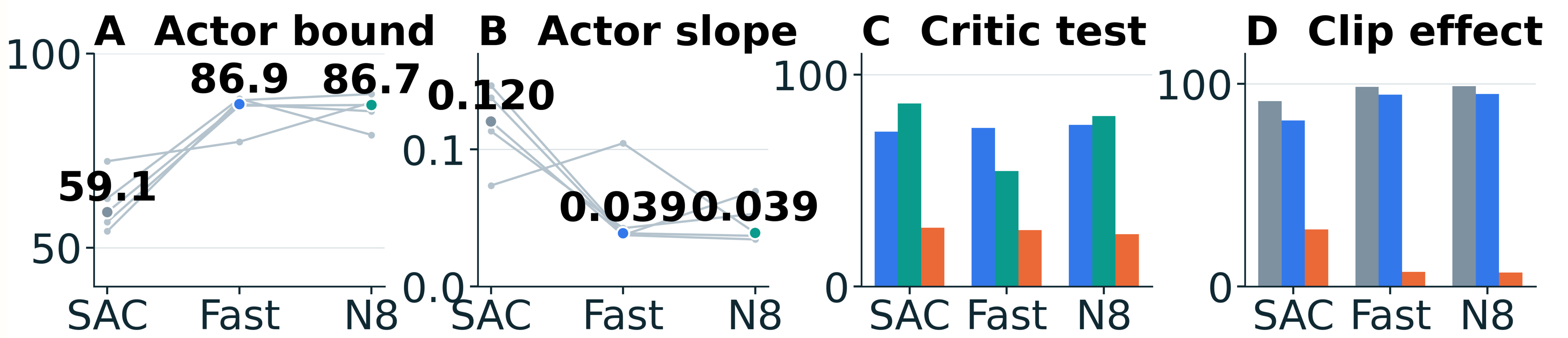
Learner-state DAgger — Reference controller labels states visited by the actor → **Priority follow-up** — Sample starts with the largest return deficit → **Local Q-search** — Score a_0 and $a_0 \pm \{0.05, 0.10\}$ with both critics

Pure RL — 100k reward transitions

FastSACN8 — Average one-step and eight-step critic losses, λ=0.5 → **Reflection** — Average the actor with its symmetry-reflected action → **Q-search** — 41 torques in [−2,2]; both critics must improve by .005

Matched target ablation inside the pure-RL networks

Only the critic target changes. Same SimbaV2, five seeds, 100k transitions, and one actor and critic update per transition. SAC uses one-step targets, Fast denotes FastSACN8 with equal one-step and eight-step targets, and SACN8 uses eight-step targets. Bound means |a| ≥ 1.99; sensitivity is mean 1−tanh²(u). Actor tests: 5,553 fixed states per seed. Critic tests: 512 fixed failures per seed. Thin traces are seeds; bold dots are medians.



C blue: ∂Q/∂a points toward the helpful action
C teal: Q ranks the helpful action above the actor
C orange: the projected critic step lowers return
D gray: actor is at the torque bound
D blue: ∂Q/∂a points beyond the bound
D orange: fraction of the proposed step retained

$a_\theta(s)=2\tanh u_\theta(s),\quad \partial a/\partial u=2[1-\tanh^2(u)]\quad a'=\Pi_{[-2,2]}\{a+0.05\,\mathrm{sign}(\partial Q/\partial a)\}$

Actor bottleneck: multistep targets raise torque-bound occupancy from 59% to 87% and reduce median output sensitivity from 0.120 to 0.039. A helpful action is the reference-controller action when its rollout return exceeds the actor's. On roughly 75% of failures, ∂Q/∂a points toward it, yet only about 7% of the proposed step survives clipping. Scoring 41 torques directly raises reflection from 94.5% to 95.9%. Both deployed pipelines use learned actors and critics without a reference query.

03 What changes the tail?

Matched intervention minus control: net classifications on 5 × 2,501 paired trials

Only component changed	Near	Task	Strict
Learner-state refit vs no refit	+29	+7	+442
Priority vs uniform sampling	+10	+2	+111
Mixed Q-search vs same actor	+26	+32	−243
Reflection vs same pure actor	+312	+10	+376
Pure Q-search vs reflection	+458	+242	+333

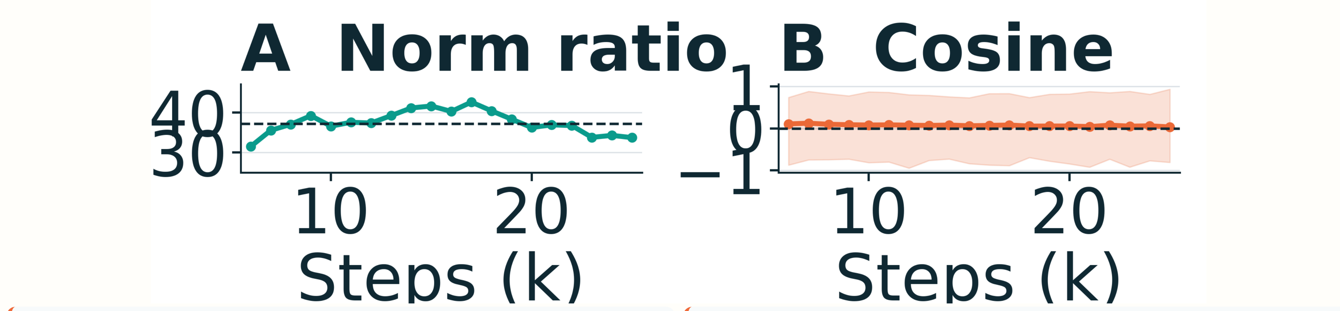
Reference-prefix repair

1,267 failures from five frozen 100k SimbaV2 FastSACN8 actors

Reference for K steps	19%	48%	91%	99%	0.6%
	K=1	K=8	K=16	K=32	shuffled 32

The same actor resumes after each prefix. Shuffling the same 32 actions reduces repair from 99.0% to 0.6%, so action order and the induced state trajectory matter jointly.

Joint-loss gradient scale



37.2× median $\|\beta g_{BC}\| / \|g_{SAC}\|$

−0.943 to +0.935 gradient cosine range

One randomly initialized SimbaV2 SACN8 seed, 25k steps, static reference anchors: joint BC plus SACN8 reaches 2,443 near-reference starts; reward-only SACN8 reaches 1,757. These are actor gradients. One coefficient was tested; normalized or projected joint losses remain open.

Critic gains can be blocked at the actor output. FastSACN8 raises critic-direction agreement, but torque-bound occupancy rises from 59.1% to 86.9% and median output sensitivity falls from 0.120 to 0.039. Clipping retains 7.1% of a nominal critic step. The 41-action search bypasses this projection and raises near-reference success after reflection from 94.5% to 95.9%.

RL plus supervision leaves 9 near-reference failures in 12,505 trials. In matched pure-RL runs, multistep actors retain about 7% of a local critic step after clipping; 41-action Q-search improves reflection from 94.5% to 95.9%.

References: dynamic programming and energy controller
SAC: Haarnoja et al., 2018 · DAgger: Ross et al., 2011
SimbaV2: Lee et al., 2025 · SACn: Łyskawa et al., 2025